

**Department of Computer Science
College of Natural Sciences
Addis Ababa University
Module Content Outline CoSc3112**

1. **Course Code:** CoSc3112
2. **Module/Course Title:** Operating System
3. **ECTS/Credit Hours:** 6/4
4. **Prerequisite:** CoSc-M1011 (CoSc1012)
5. **Course Objectives & Competences:**

Upon completing the module, the student would be able to:

- Explain the basic structure and functioning of operating system;
- Understand problems related to process management and synchronization;
- Explain the cause and effects related to deadlocks;
- Analyse circumstances that lead deadlock in operating systems;
- Understand the basics of memory management, and
- Understand the structure of modern file-systems.

6. **Module Outline**

Chapter 1: Introduction to Operating Systems

- 1.1 What is an operating system?
- 1.2 History of operating systems with respect to computer generations
- 1.3 Operating system services
- 1.4 Types of operating systems and operating system structure

Chapter 2: Processes and Process Management

- 2.1 The process concept
- 2.2 The threads concept
- 2.3 Inter-process communication
- 2.4 Process scheduling, Scheduling Algorithms, Deadlocks

Chapter 3: Memory Management

- 3.1 Basic memory management
- 3.2 Multiprogramming with variable partitions- swapping
- 3.3 Virtual memory
- 3.4 Shared pages- a design issue
- 3.5 Segmentation

Chapter 4: File Systems

- 4.1 Introduction to file systems
- 4.2 Files
- 4.3 Directories or folders
- 4.4 File system implementation

Chapter 5: Input/output Management

- 5.1 principles of I/O hardware
- 5.2 principles of I/O Software
- 5.3 I/O software layers

Chapter 6: Security and Protection

6.1 Overview of OS Security

6.2 Threats, intruders, accidental data loss, encryption, user authentication

6.3 Other security issues: Trojan horse, viruses, worms, Logic bomb

7. Evaluation

Tests: 40%

Final Examination: 60%

8. Text Book

- A.Tanebaum:modern Operating Systems,2nd edition,prence-hall,2001
- W.Stallings: Operating Systems: internals and Design principles,5th edilion,prenticeHall,2005.
- A Silberschhatz,P.B.Galvin and G,Gangne:Operating system Concepts,6th edition,John Wily & Sons.2001..